Bayesian Inference I

Andaleeb Hassan

```
library(tidyverse)
library(gridExtra)
```

problem 1

$$\begin{aligned} Prior &: \theta \sim Beta(a, b) \\ Sampling\ model &: y|\theta \sim Binomial(n, \theta) \\ Posterior &: \theta|y \sim Beta(a + y,\ b - n + y) \end{aligned}$$

```
x <- seq(0,1,length.out = 10000)
y11 <- dunif(x)
y12 <- dbeta(x,2,5)
m1 <- ggplot()+
  geom_line(aes(x,y11))+
  geom_line(aes(x,y12))+
  theme_bw()

y21 <- dbeta(x,3,2)
y22 <- dbeta(x,4,6)

m2 <- ggplot()+
  geom_line(aes(x,y21))+
  geom_line(aes(x,y22))+
  theme_bw()

y31 <- dbeta(x,1,1)
y32 <- dbeta(x,21,81)

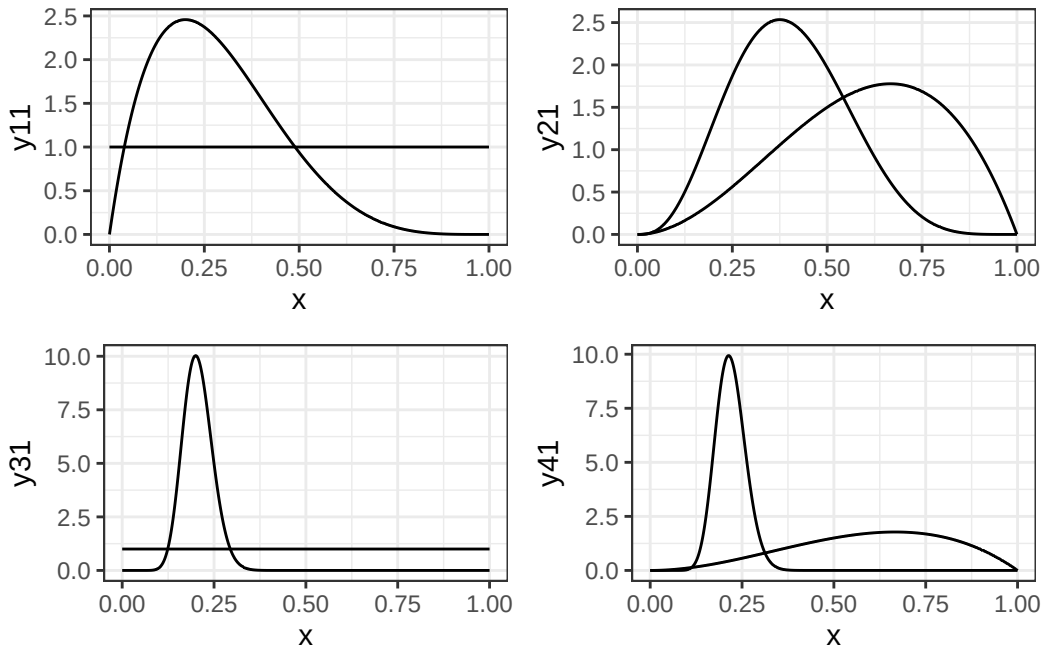

m3 <- ggplot()+
```

```
  geom_line(aes(x,y31))+
  geom_line(aes(x,y32))+
  theme_bw()

y41 <- dbeta(x,3,2)
y42 <- dbeta(x,23,82)

m4 <- ggplot()+
  geom_line(aes(x,y41))+
  geom_line(aes(x,y42))+
  theme_bw()

grid.arrange(m1,m2,m3,m4,nrow =2)
```



problem 2

Sensitivity analysis:
Parameters of beta distribution a and b in terms of $\theta_0 = \frac{a}{a+b}$ and $\eta_0 = a + b$, so that $a = \theta_0\eta_0$ and $b = (1 - \theta_0)\eta_0$. For each $\theta_0 \in \{0.1,\ 0.2, \dots,\ 0.9\}$ and $\eta_0 \in \{1,\ 2,\ 8,\ 16,\ 32\}$.

Find:
1. Corresponding a, b values.
2. $Pr(\theta > 0.5 \mid \sum Y_i = 57)$

```
theta_not  <- seq(.1,.9,by =.1)
eta_not    <- c(1, 2, 4, 8, 16, 32)

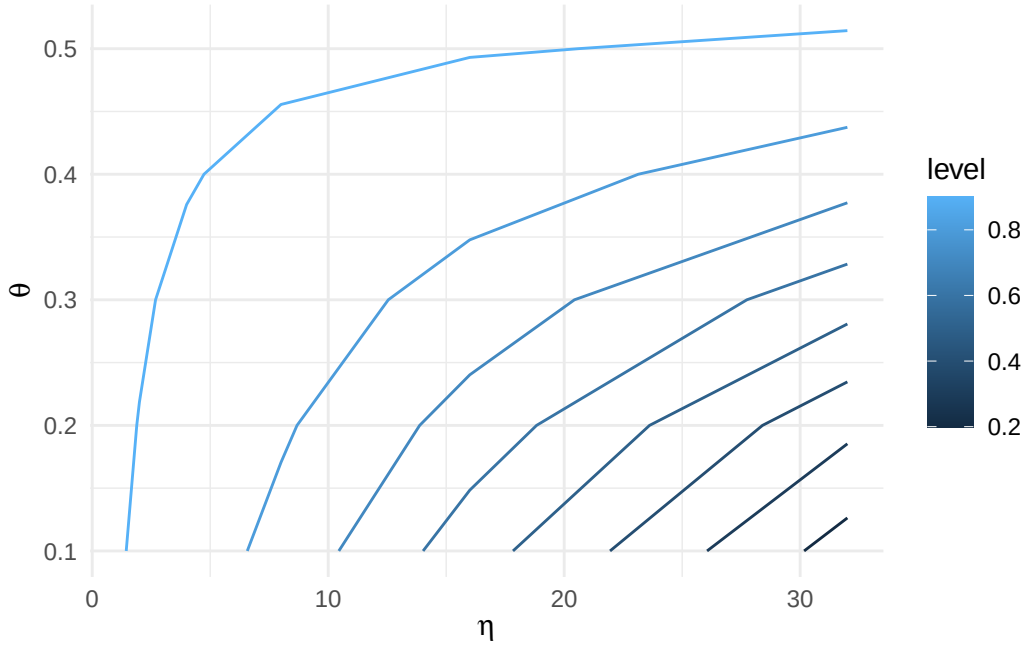
data <- tibble( theta = theta_not)

data <- data %>%
  expand_grid(tibble(eta = eta_not))


data <- data%>%
  mutate(density =  pbeta(.5,theta*eta+57,eta*(1-theta)-57+100,
                          lower.tail = FALSE ),
         a = theta*eta,
         b = eta*(1-theta))



ggplot(data =data)+
  geom_contour(
    aes(y= theta, x = eta,
                  z = density,
        color = after_stat(level)))+
  labs(y = expression(theta),
       x = expression(eta))+
  theme_minimal()
```


0.5
0.4
0.3
0.2
0.1
θ
0
10
20
30
η
level
0.8
0.6
0.4
0.2